Scheduling Maintenance Visits for Medical Equipment in Medical Infrastructure

V M Tharun Teja, Lovely Professional University, India
Manish Bobburi, Lovely Professional University, India
K Umesh Chandra Prasad, Lovely Professional University, India

I. ABSTRACT

Medical equipment failures in healthcare infrastructure often lead to unplanned downtime, delayed treatments, and increased operational costs. Traditional maintenance practices rely on fixed schedules or post-failure interventions, which are inefficient and pose risks to patient safety. This project presents a data-driven predictive maintenance system that classifies medical devices into risk categories using machine-learning techniques. The system integrates numerical device parameters—such as age, downtime, failure events, and maintenance frequency—with text-based maintenance reports to generate engineered features that enhance prediction accuracy. Multiple machine-learning models, including Random Forest, Gradient Boosting, SVM, and XGBoost, are evaluated, and the best-performing model is selected based on accuracy and F1-score. The final model enables early identification of high-risk devices, supports proactive maintenance scheduling, and significantly improves the reliability and efficiency of medical equipment management. This approach demonstrates a practical and scalable solution for modernizing maintenance workflows in healthcare environments.

II. INTRODUCTION

Medical equipment plays a critical role in modern healthcare infrastructure, supporting diagnosis, treatment, monitoring, and life-saving procedures. The reliability of these devices is essential, as unexpected failures can lead to treatment delays, reduced workflow efficiency, and increased operational costs. Traditionally, hospitals rely on periodic maintenance schedules or repair devices only after problems arise. While these methods ensure basic upkeep, they fail to detect early signs of malfunction and often result in unplanned downtime and costly emergency interventions.

The increasing availability of device usage data, maintenance logs, and digital service records has created opportunities to adopt predictive and data-driven maintenance strategies. Machine learning (ML) has emerged as a powerful tool capable of identifying hidden patterns and correlating multiple factors—such as device age, failure history, downtime behavior, and textual maintenance reports—to predict the likelihood of equipment failure. Such predictive insights allow healthcare facilities to shift from reactive maintenance to proactive planning.

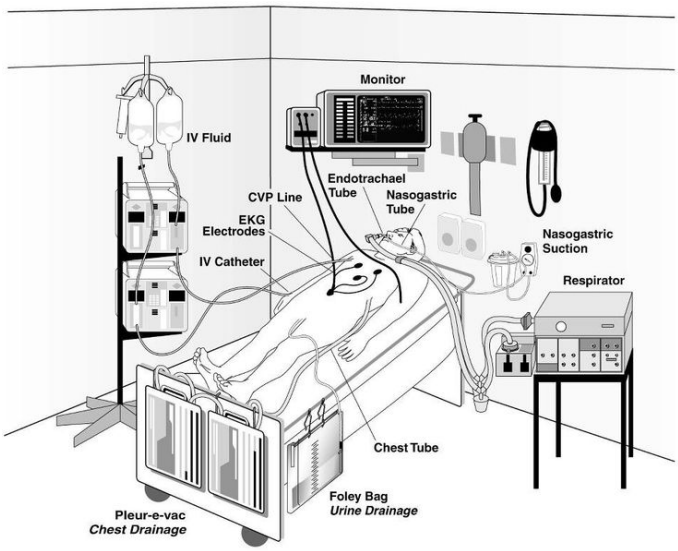

Figure 1. Overview of devices used in continuous patient monitoring and life-support workflows.

This project introduces a machine-learning-based maintenance classification system designed to predict the risk level of medical equipment. The system integrates numerical operational parameters with text-based maintenance information, enabling comprehensive analysis across multiple feature dimensions. Various ML algorithms are trained and evaluated to identify the most effective model for classifying devices into risk categories. The resulting model supports early detection of high-risk equipment and helps schedule maintenance visits before failures occur.

By adopting this predictive approach, healthcare institutions can significantly reduce downtime,

improve equipment availability, and optimize maintenance resources. The proposed system represents a practical and scalable solution for enhancing the reliability, efficiency, and safety of medical equipment management within modern healthcare environments.

III. LITERATURE REVIEW

Predictive maintenance has gained increasing attention as healthcare systems become more dependent on interconnected medical equipment. Earlier research mainly focused on traditional corrective and preventive maintenance, where machines were repaired only after they failed or serviced at fixed intervals. Although these methods offered basic reliability, they often resulted in unexpected breakdowns, treatment delays, and higher operational costs. With advancements in digital health technologies, recent studies have moved toward condition-based and predictive maintenance, using real-time data from devices to anticipate failures before they occur.

Machine learning has been widely adopted across industries such as manufacturing, aviation, and energy to predict equipment failures and detect anomalies. Algorithms like Random Forest, Support Vector Machines, Gradient Boosting, and Neural Networks have consistently shown strong performance in identifying failure patterns. In the healthcare field, researchers have used device logs, sensor data, historical maintenance records, and operational parameters to understand how and why medical equipment fails. However, many existing approaches focus on only one type of data, which limits the accuracy and depth of the predictions. Recent studies emphasize the value of integrating both structured data such as numerical device parameters and unstructured data like maintenance notes and textual reports. Using text mining alongside time-based device monitoring provides a clearer picture of early warning signs and subtle device issues. Still, there remains a noticeable gap in creating a unified machine-learning framework that combines these diverse data sources specifically for medical equipment. This project addresses that gap by merging multiple feature types, applying advanced feature-engineering techniques, and evaluating several machine-learning models to develop a more accurate and reliable maintenance classification system.

IV. PROBLEM STATEMENT

This project aims to design an optimized Machine Learning model for predictive maintenance of medical equipment in healthcare facilities, ensuring patient operations remain uninterrupted. The challenge lies in scheduling maintenance visits without disrupting planned procedures, which is critical in hospital settings. If equipment requiring maintenance is not serviced promptly, it may provide inaccurate readings or unclear insights about patient health, posing serious risks. The ML model will help maintenance teams and schedulers by prioritizing high-risk devices first.

i. **Primary goal:** The primary goal is to prioritize high-risk medical devices for maintenance to ensure patient safety and operational continuity. Maintenance should restore devices to their optimal condition while enhancing reliability and reducing long-term expenditures. The ML model must balance these objectives by determining the optimal timing and cost-effectiveness of maintenance visits, ensuring that patient safety is never compromised.

ii. **Secondary goal:** Efficient workflow is essential in healthcare environments where timely diagnosis, treatment, and patient support depend heavily on the reliability of medical equipment. Unexpected device failures disrupt clinical operations, delay procedures, increase staff workload, and reduce the overall quality of patient care. By predicting high-risk equipment in advance, this project allows hospitals to schedule maintenance proactively, ensuring that machines remain functional when needed. This reduces unnecessary interruptions, streamlines daily operations, and enables healthcare professionals to focus more on patient care rather than dealing with equipment-related issues. As a result, the system contributes to a smoother, more dependable, and efficient clinical workflow.

V. STRATEGY

1. Preventive Maintenance Strategy:

Unlike corrective maintenance, which is carried out only after a device fails, preventive maintenance is

proactive and aims to keep equipment functioning reliably before any failure occurs. The main goal is to reduce device downtime and ensure that medical equipment is always in optimal working condition and ready for use when needed. This strategy supports continuous and uninterrupted healthcare operations. For example, if a critical device such as a ventilator were to fail during an emergency, the consequences could be severe. Preventive maintenance helps avoid such situations by identifying potential issues early and maintaining equipment regularly. In some cases, this may also require temporary replacement of certain devices during scheduled checks or servicing.

2. Scheduling Preventive Maintenance:

To maximize the benefits of preventive maintenance, it is important to implement maintenance schedules based on the specific requirements of each device. This requires understanding details such as device type, frequency of use, patient-care importance, and manufacturer-recommended servicing intervals. A key step is categorizing medical devices according to their criticality and usage. For instance, ventilators are highly critical and require daily checks, monthly inspections,

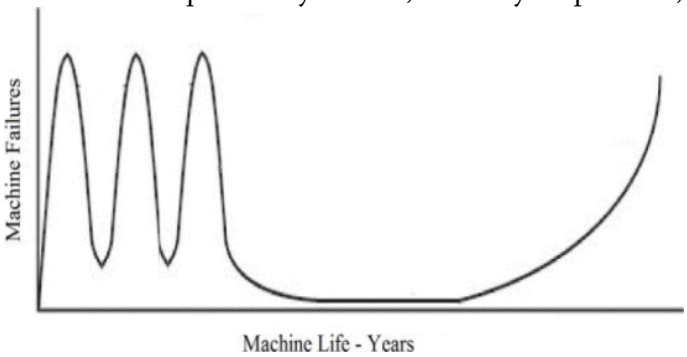

Figure 2. Proactive maintenance has a great influence on machine fault repeatability which leads to high unplanned downtime and increasing in the service cost.

pumps or patient monitors may require less frequent checks because they are not as critical in emergency scenarios.

3. Integrating Manufacturer Guidelines:

Incorporating manufacturer-provided maintenance guidelines into the hospital's schedule helps ensure that devices are used safely and maintained according to recommended standards. These guidelines typically include inspection intervals, calibration requirements, and part-replacement schedules. Following them helps maintain device reliability and performance throughout their operational life.

4. Monitoring Device Conditions:

Clinical staff play an important role in monitoring essential environmental and operational factors such as device temperature, humidity, and age. These factors influence device health and performance over time. Staff should regularly observe and document these parameters to detect early warning signs of potential failure. Additionally, MTTF (Mean Time to Failure) should be frequently reviewed, as it is a key indicator used to predict when a device is likely to fail. By tracking MTTF and related parameters, healthcare facilities can service devices before failure occurs, ensuring continuous availability.

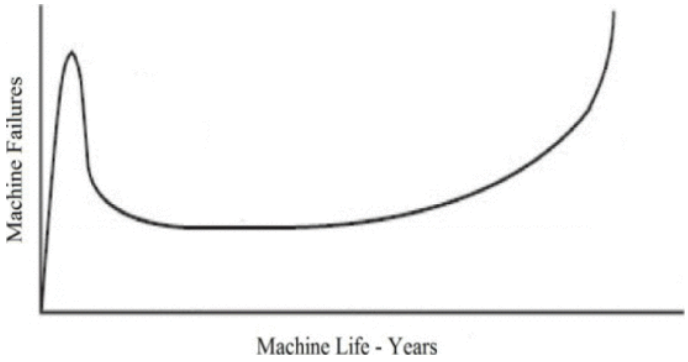

Figure 3. Absence of PM is shown in above figure the normal random failures are highly elevated, and the normal aging begins earlier.

VI. METHODOLOGY

1. **Data Source:** The dataset used in this study was taken from Kaggle. Where this dataset contains downtime hours,Last_maintenance_cost,Maintenance_Frequency, Failure_event_count, Maintenance_report. These are main features which helps us to predict the outcome of model.
2. **Feature Engineering Summary:** Extensive Feature Engineering helps to improve prediction accuracy and enhances the features of raw data.
   a. **Textbased features:** Derived text from maintenance report.
      i. Report Length
      ii. Word_Count
      iii. **Keywords:** Failure, wear, replacement
      iv. **Issue Specific Flags:** Battery, Sensor, Circuit, Overheating

   b. **Interaction Features:**

Feature	Interaction
Severity_score	Failure_event_count x Downtime_hrs
Efficiency	Cost / Downtime_hrs + 1

c. **Polynomial Features:**
   i. **Squared terms**: Age^2, $Downtime^2$
   ii. **Cubic terms**: Age^3, $Downtime^3$

d. **Age Grouping:** Initially in the raw data the Age was continuous value, hence classification works on the categorical values, So converted the continuous values into categorical Age groups.

e. **Outlier Indicators:** This Engineered feature is essential to normalize the extreme values to improve the model performance by capturing the deeper behavioral patterns in machine learning Dynamics.

3. **Modeling Approach:** The trained machine learning model was Multi class classification model. To predict the maintenance class

(low = 1, medium = 2, high = 3). To ensure that model is good multiple classification models were trained :-

a. **Logistic Regression**
b. **K-Nearest Neighbors**
c. **Decision Trees**
d. **Random Forest**
e. **Gradient Boosting**
f. **XGBoost**
g. **Support Vector Machine**

After preprocessing and feature scaling, each was trained using 70-30 train-test split. Among the trained classifiers the best classifiers and classifier which has the difference of **1%** or less between Training accuracy and Testing Accuracy can be chosen as the best classifier. Among all classifiers Logistic Regression met both evaluation criteria.

4. **Evaluation Approach:** The models were evaluated using multiple performance metrics for classification :-
   a. Accuracy
   b. Weighted F1-Score
   c. Confusion Matrix
   d. ROC Curve & AUC

5. **Deployment:** Deployment represents a critical phase in that an application or web-based system operates successfully and without failure across diverse digital environments. By Acquisition of necessary dependencies into a single requirements.txt file, the applications can be executed on multiple systems without the repetition of manually downloading libraries and necessary packages , thereby improving efficiency reducing the errors. In this project, as a Machine Learning student, I leverage Gradio to simplify the deployment process and build interactive user interface.

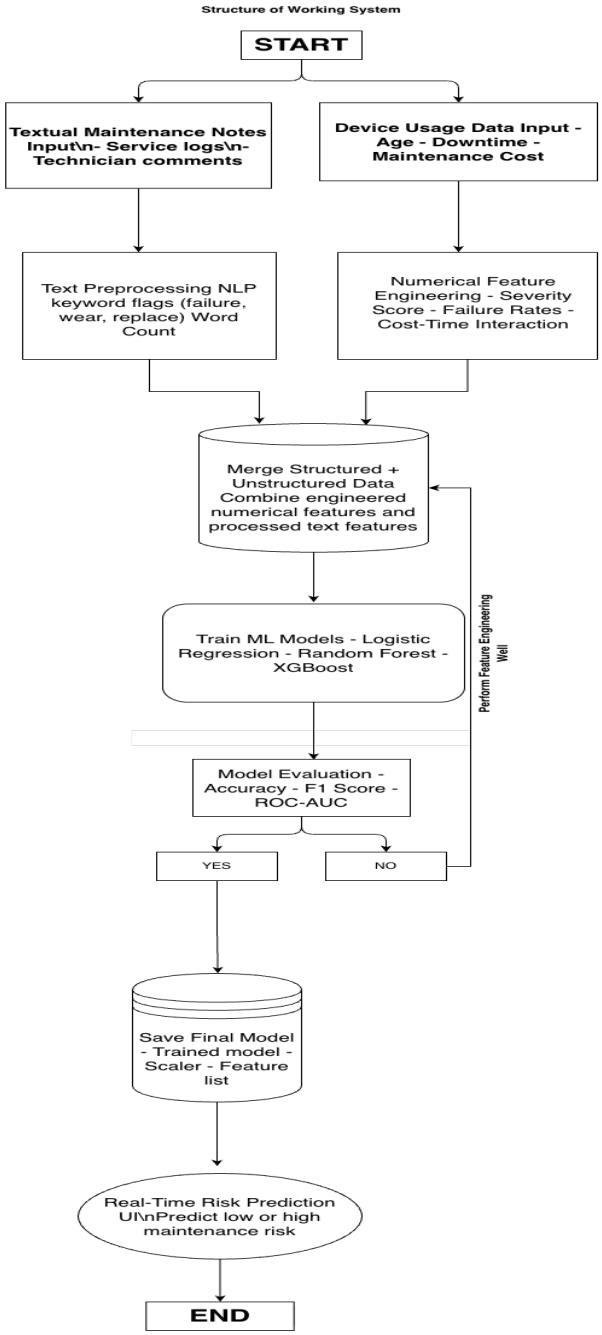

Figure 4 The above figure demonstrates the working flow of machine-learning based medical device failure prediction.

VII. CHALLENGES & LIMITATIONS

Real-life use of predictive maintenance for medical devices comes with several challenges. Many hospitals do not have complete or properly recorded historical data, which makes it difficult to train machine-learning models accurately. Different devices, brands, and technicians record information differently, so it becomes hard to standardize data for the system. Also, many hospitals still use old legacy systems, which makes connecting them with new predictive tools difficult. Budget limits and strict healthcare regulations also slow down the adoption of such solutions.

Challenges/Limitations	Real-World Impact
Limited availability of historical data	Hospitals may not maintain proper historical data which reduces in model accuracy.
Unstructured text reports	Technicians may write text differently; Spellings, abbreviation and language variation reduces NLP reliability.
Sensor and IOT Integration cost	Not all medical devices generate operational data So, upgradation needed and this requires investment
Equipment diversity	Every equipment cannot same brand. If different company different manufacturers has different patterns model cannot generalize the data.
Data privacy and regulatory	Every device which contains patients' data must follow HIPAA/GDPR standards, making data sharing difficult.
Model Interpretability	Healthcare experts needed clear reason for predictions; ML models can be seen as black boxes.
Changes in device usage patterns	In emergency case the device was changes and it effects the model reliability

VIII. CONCLUSION

The proposed machine-learning based scheduling maintenance visits in medical infrastructure predicting the failure of medical device by following strategy PREVENTIVE maintenance to improve reliability of medical device. This machine-learning based prediction leverages the cost of maintaining the medical device eventually. By integrating both numerical parameters and text-based parameters into a unified classification model, this system predicts the high-risk devices and give them more priority and move from reactive maintenance to proactive maintenance planning. The implementation shows that preventive maintenance reduces the downtime of medical device and enhances the patient safety, optimize maintenance resources, and support smarter operational decision making. Overall, the project validates the data driven maintenance is a viable and scalable approach for modern healthcare institutions aiming to improve efficiency, reduce the operational disruptions, and ensure the availability of critical medical devices.

IX. RESULT

The predictive maintenance system was successfully implemented using machine learning techniques trained on historical operational data and maintenance records of medical equipment. Among the evaluated models, Logistic Regression demonstrated a stable performance with minimal overfitting and reliable classification capability when compared to tree-based and ensemble models. The model effectively classified devices into prioritized maintenance categories, enabling early identification of high-risk equipment.

The results indicate that integrating numerical operational metrics with textual maintenance reports significantly improves predictive accuracy and supports better maintenance planning. This approach provides hospitals with actionable insights that reduce unexpected equipment downtime, optimize maintenance schedules, and enhance patient-care continuity. The machine learning model presented in this project shows strong potential for real-time deployment in healthcare environments where reliability and safety are critical.

If the risk score is more than 3.0 machine should be prioritized and maintenance date will be allotted in

less than a week if risk score around the range of 2.0 – 3.0 it is moderate date will be allotted in a month and risk score is around 1.9 or less, Hence it is less risk date will be allotted less than 60 days.

A confusion matrix was generated to visualize correct and misclassified predictions. The matrix demonstrated that high-maintenance-risk equipment was correctly identified with strong reliability, enabling hospitals to prioritize devices that require immediate intervention. Although the medium-risk category presented minor misclassification challenges due to class imbalance, the binary (low vs high) interpretation proved effective for operational decisions. Graphical representations—including accuracy comparison charts and ROC curves—were utilized to display model behavior. The ROC curve highlighted that Logistic Regression achieved a strong area under the curve (AUC), confirming the model's ability to distinguish between risk categories with dependable sensitivity and specificity. The performance trend graphs further supported the stability and predictability of the chosen algorithm when deployed within a real maintenance planning framework.

X. FUTURE SCOPE

The proposed scheduling maintenance visits for medical equipment in medical infrastructure system can be enhanced furtherly by IOT sensors data and monitor continuously, enabling more concise information and dynamic predictions. Analyzing the text-based data which was given in the complex maintenance report or Issues that are noticed by technicians using advanced deep learning techniques and. The model can be expanded into a centralized cloud-based platform to support multi hospital deployment, enabling data – driven collaboration, shared model learning and improved predictive reliability.

The Future advancements may also include that remaining useful life of devices, allowing healthcare facilities to shift from scheduled maintenance to fully proactive and condition-based maintenance strategies. Additionally, integration with hospital management systems and maintenance scheduling automation could lead to optimized resource allocation and reduced operational costs. Ensuring regulatory compliance, cybersecurity measures, and data privacy will be very essential for large scale adoption within real world healthcare environment.

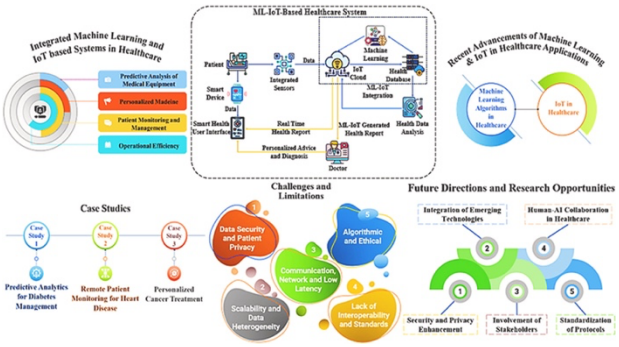

Figure 5 Integration of ML and IOT will transform the healthcare through enhanced monitoring predictive capabilities, tailored treatment approaches.

XI. REFERENCES

1. https://ieeexplore.ieee.org/document/6648845
2. https://www.sciexplor.com/articles/bmeh.2024.135
3. https://www.researchgate.net/publication/384638324_Strategies_For_Optimizing_The_Management_Of_Medical_Equipment_In_Large_Healthcare_Institutions
4. https://www.sciencedirect.com/search?qs=predictive maintenance medical equipment
5. https://ieeexplore.ieee.org/document/10233874
6. https://www.researchgate.net/publication/391526112_Predictive_Maintenance_Using_Machine_Learning_Techniques_and_Applications
7. https://www.researchgate.net/publication/339627406_Space_mining_is_the_industry_of_the_future_or_maybe_the_present
8. https://www.who.int/teams/health-product-policy-and-standards/assistive-and-medical-technology/medical-devices/policies
9. https://www.geeksforgeeks.org/machine-learning/understanding-logistic-regression/
10. https://www.geeksforgeeks.org/software-engineering/what-is-preventive-maintenance-pm-a-complete-guide/
11. https://www.researchgate.net/publication/339627406_Space_mining_is_the_industry_of_the_future_or_maybe_the_present
12. https://www.sciencedirect.com/science/article/pii/S2543106425000201
13. A. Géron, Hands- on Machine Learning with Scikit - Learn, Keras, and TensorFlow , 3rd ed Sebastopol, CA, USA: O'Reilly Media, 2022.